

ASM-HEMT Model Release Notes

Version 101.5.0 vs Version 101.6.0

Prepared by: Sourabh Khandelwal

1. Summary

ASM-HEMT v101.6.0 introduces functional and modeling updates relative to v101.5.0, including enhancements to gate-current modeling (new gatemod=4 formulation), diode current computational stability improvement, and cleaner code with several unused variables and equations streamlines. The changes remain backward compatible.

2. High-Level Functional Changes

2.1 Gmin for gate diode

Change	v101.5.0	v101.6.0	Impact
Added global Gmin parameter	Not present	Added (<code>Gmin = \$simparam("gmin",0.0)</code>)	Improves convergence

2.2 Gate-Current Modeling Enhancements

Feature	v101.5.0	v101.6.0	Impact
IDIO3 (gatemod=3)	Present	Present	No change
New IDIO4 (gatemod=4)	Not present	Added	More accurate reverse-bias leakage
Reverse-bias smoothing	Basic	Improved	Better numerical stability

3. Other Improvements

This version has streamlined equations and unused variables have been removed. A few output variables had incorrect mult_q factor, these have been fixed in this version.